EFY TESTED
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AUTOMATIC WIPER CONTROL
For Three/Four Wheelers

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The integration of a 555 timer and L293D motor driver in an automatic wiper control device presents a practical and efficient solution for enhancing driver safety and convenience. This system, with its simplicity and cost-effectiveness, can be a valuable addition to both three- and four-wheelers. Fig. 1 shows the author's prototype on a breadboard.

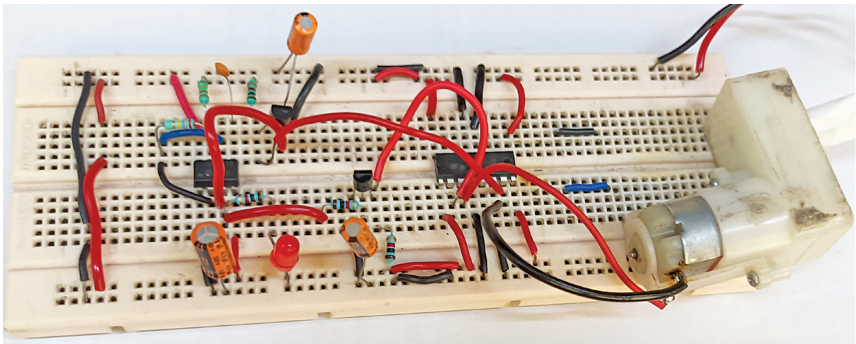

Fig. 1: Author's prototype on breadboard

Circuit and working

The circuit diagram of the automatic wiper control for three/four-wheelers is shown in Fig. 2. It is built around a 12V battery, timer 555 (IC1), motor driver L293D (IC2), 5V voltage regulator 7805 (IC3), and a few other components. LED1 acts as a power-on indicator when switch S1 is turned on, activating the wiper control system.

The heart of the device is the 555 and L293D. The 555 timer (IC1) is a highly stable integrated circuit used for generating precise time delays or oscillation. It is wired as an astable multivibrator. In a traditional astable multivibrator circuit, a duty cycle of 50% or less cannot be generated. To achieve a duty cycle of less than 50%, diode D1 is added across resistor R3, resulting in an output frequency of approximately 0.03Hz. The output of IC1 is given to transistors T1 and T2 to provide pulses, which are used as the inputs of IC2 to drive the L293D.

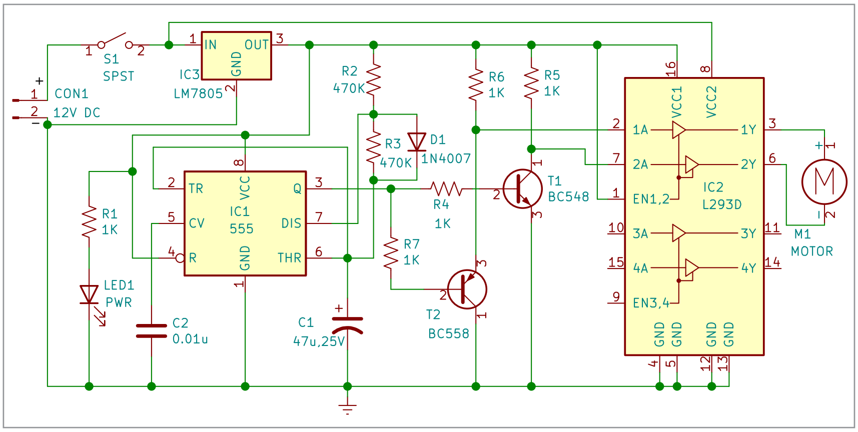

Fig. 2: Circuit diagram

L293D PIN LOGIC FOR MOTOR DIRECTION

Pin1 (EN1,2)	Pin2 (1A)	Pin7 (2A)	Pin3 (1Y)	Pin6 (2Y)	Motor Rotation
H	H	L	H	L	Clockwise
H	L	H	L	H	Anti clockwise

The L293D is a quadruple high-current half-H driver designed to provide bidirectional drive currents of up to 600mA at voltages from 4.5V to 36V. It is ideal for controlling the wiper motor, allowing for forward and reverse motion essential for wiper operation.

The output pin 3 of IC1 is connected to one end of two resistors, R4 and R7. The other end of resistor R4 is connected to the base of the NPN transistor T1 to provide a signal at pin 7 of the L293D, while the other end of resistor R7 is connected to the base of the PNP transistor T2 to provide a signal at pin 2 of the L293D.

Normally, when there is no rain, switch S1 should be off to disable